Hibernate Environment Set Up

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1 Basic Steps to follow

Follow the below mentioned instructions to successfully set up the initial environment for Hibernate

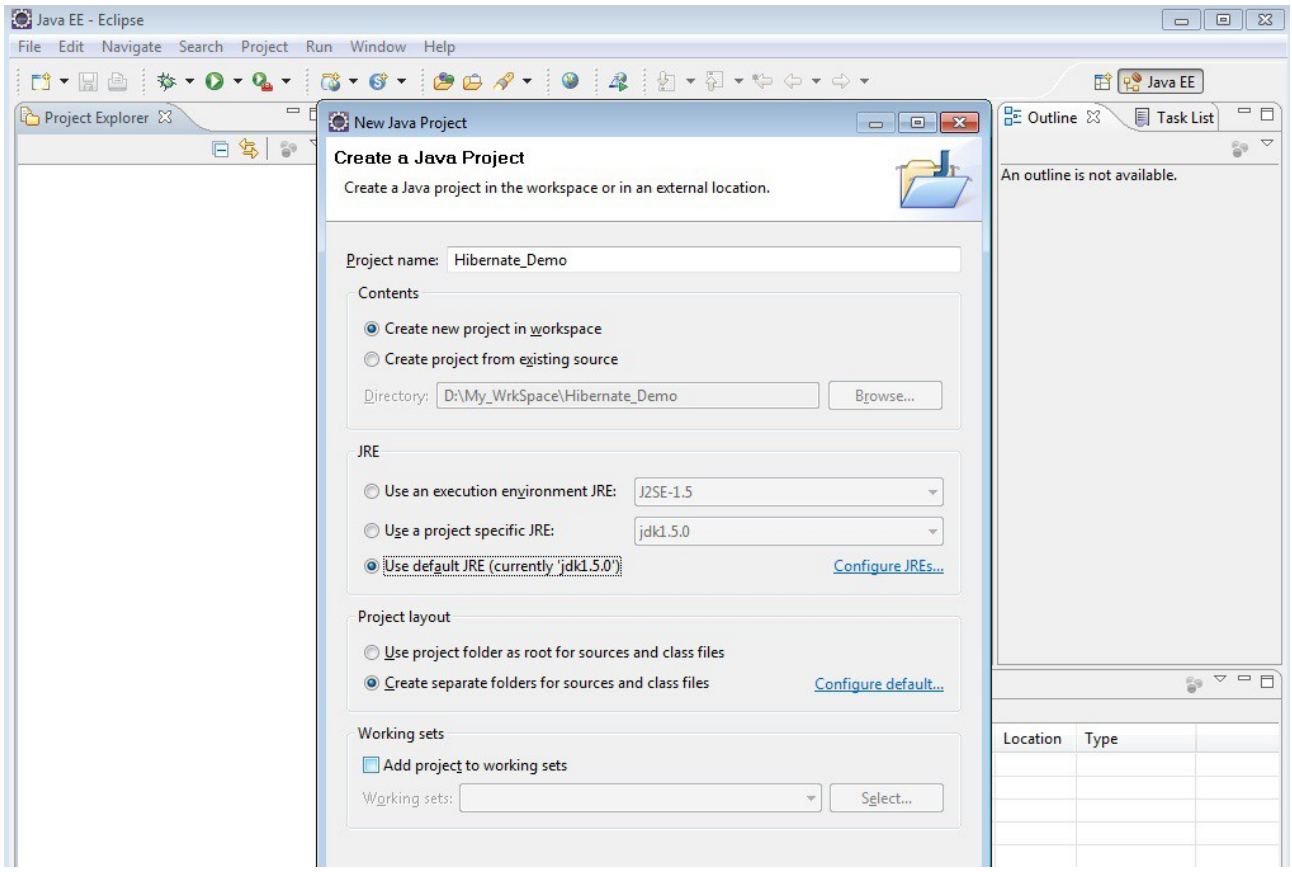
- ⇧ Create Sample JAVA Project as Hibernate_Demo
- ⇧ Import neccessary jar files as per Hibernate version(Hibernate 3.0/Hibernate 4.0 as attached in this doc)
- ⇧ Add **hibernate.cfg.xml** file in the class path of Hibernate_Demo Project
- ⇧ Prefill configuration entries in **hibernate.cfg.xml** as per your Project(e.g. DB Credentials a109core/a109core)
- ⇧ Append POJO mappings in **hibernate.cfg.xml** as explained in O/R Mappings
- ⇧ Create Test Main Class with initial coding of valid SessionFactory and Session Object
- ⇧ Compile and Run Test Main Class

2 List of mandatory jar files to run sample hibernate project

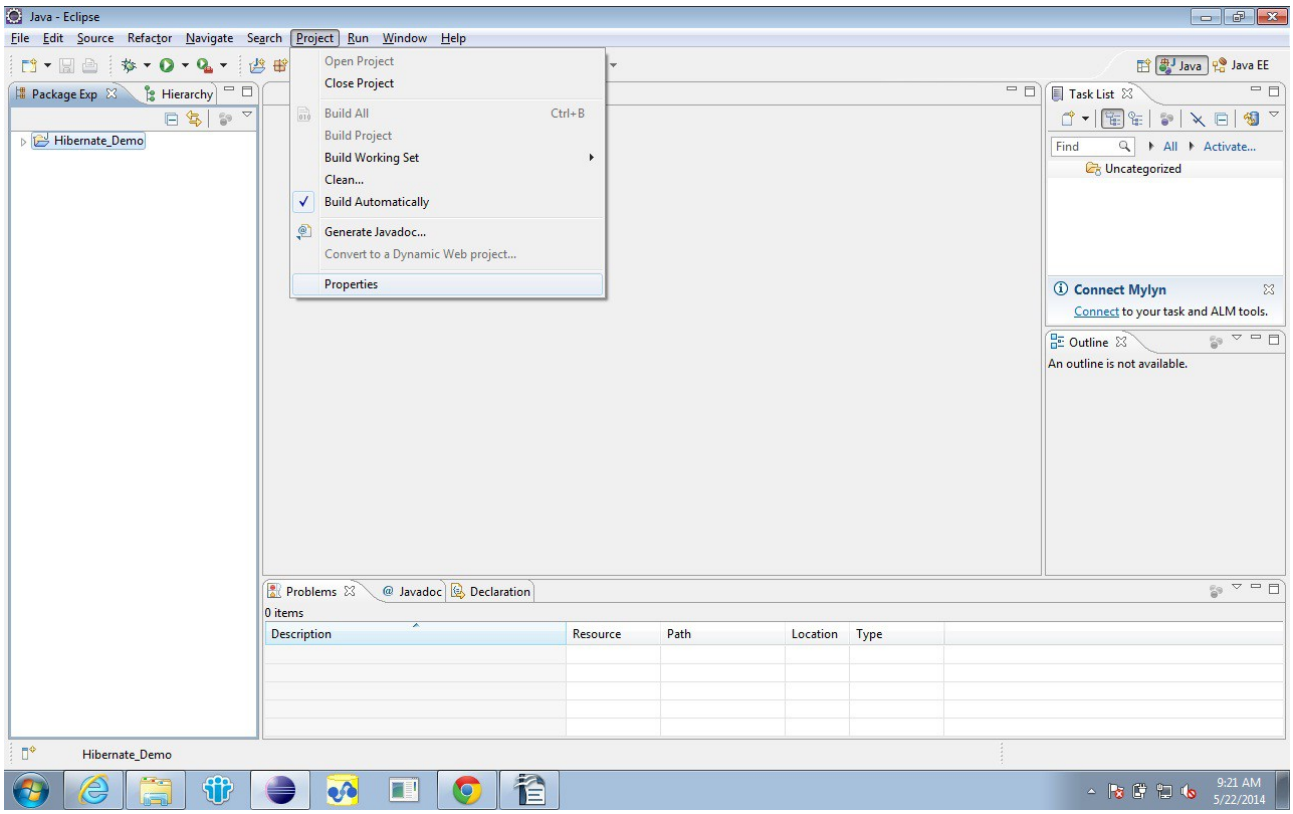
1. antlr-2.7.6
2. c3p0-0.9.1
3. classes12
4. commons-collections-3.1
5. dom4j-1.6.1
6. ehcache-core-2.4.3
7. hibernate-commons-annotations-4.0.1.Final
8. hibernate-core-4.0.1.Final
9. hibernate-core-4.1.10.Final
10. hibernate-ehcache-4.1.10.Final
11. hibernate-jpa-2.0-api-1.0.1.Final
12. hibernate-testing
13. javassist-3.12.0.GA
14. jbosscache-core-3.2.1.GA
15. jboss-logging-3.1.0.GA
16. jboss-transaction-api_1.1_spec-1.0.0.Final
17. jta-1.1
18. oscache-2.1
19. proxool-0.8.3
20. slf4j-api-1.6.1
21. swarmcache-1.0RC2

3 Step by Step execution

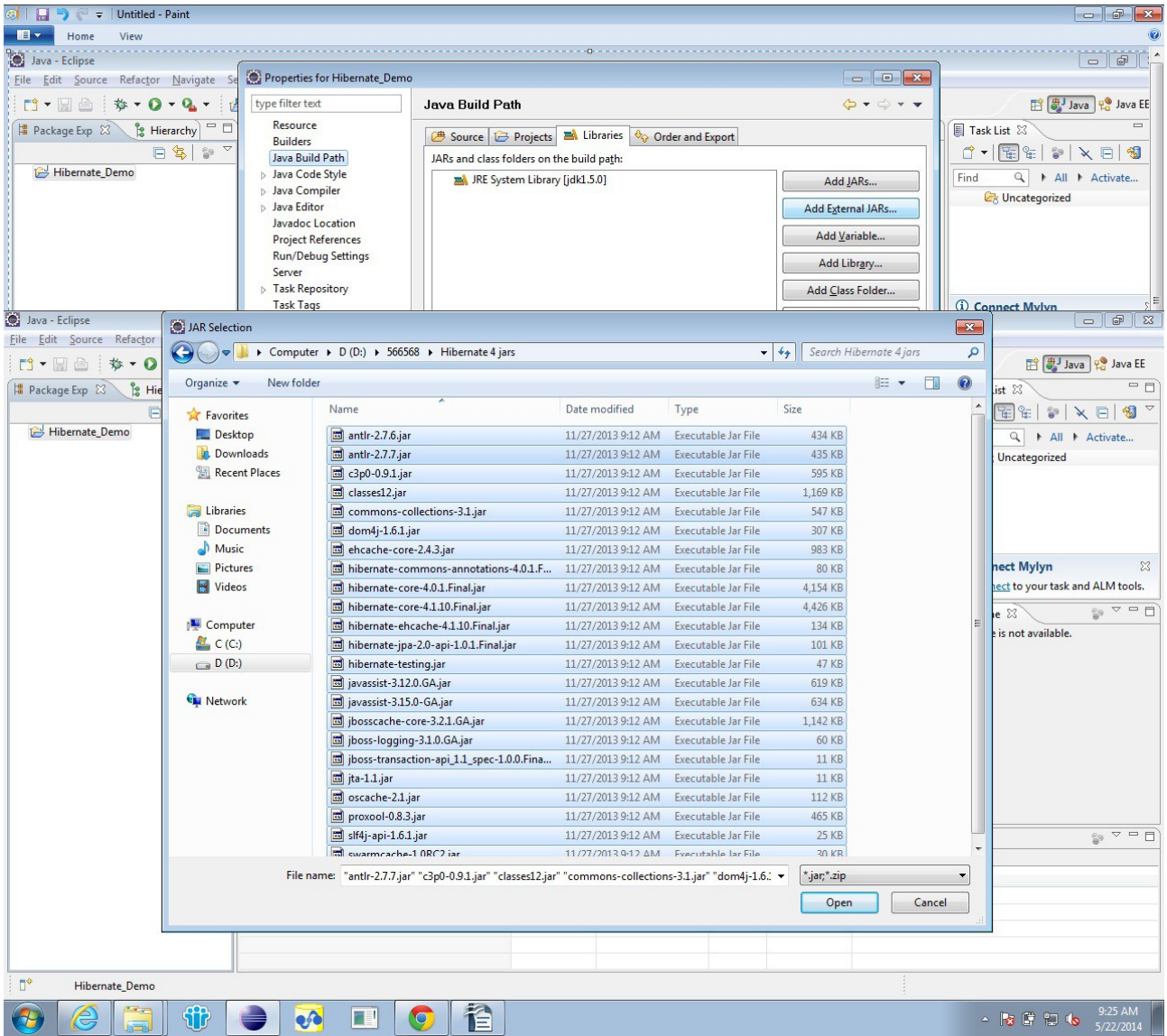
a) Create Sample JAVA Project as Hibernate_Demo



b) Go to Project(Tab) ---> Properties

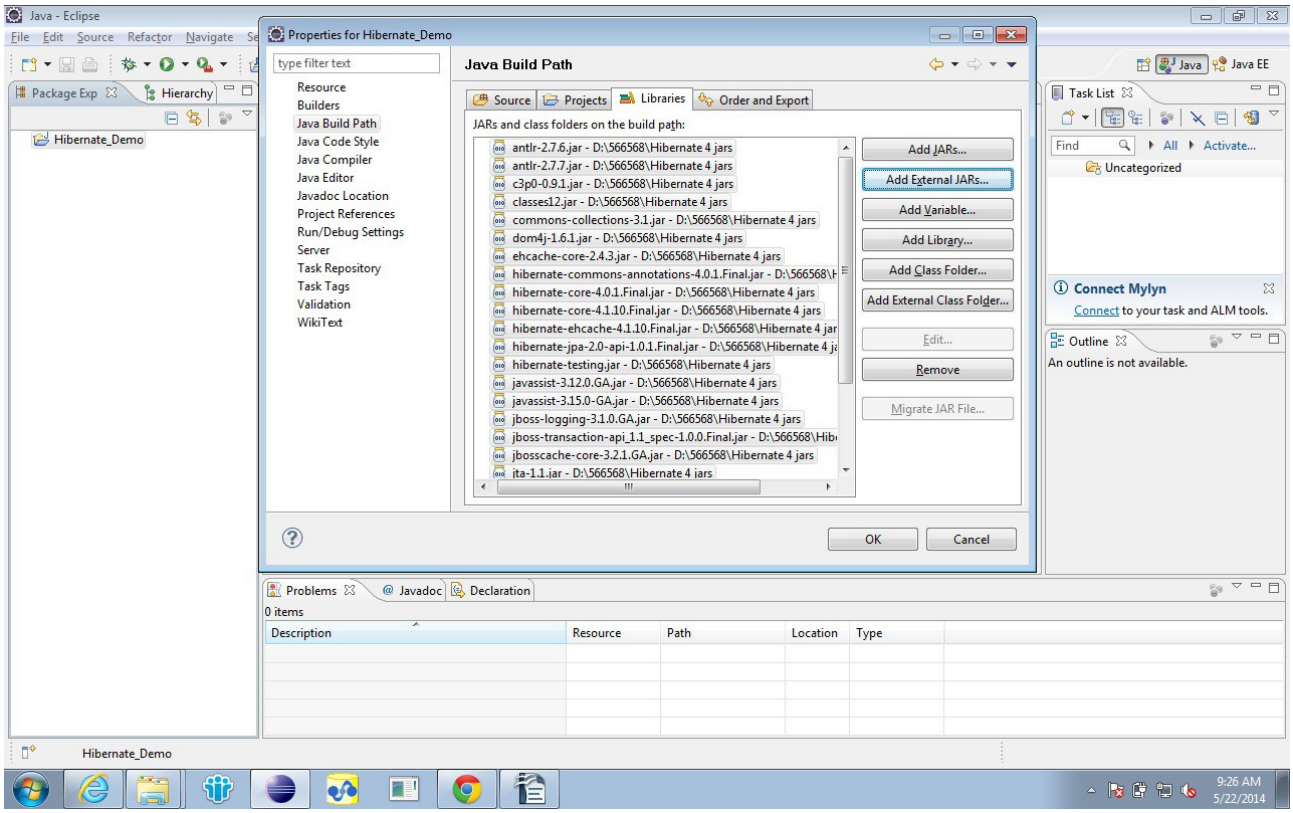


c) <u>Click on JAVA Build Path--> Libraries --> Add External Jars</u>

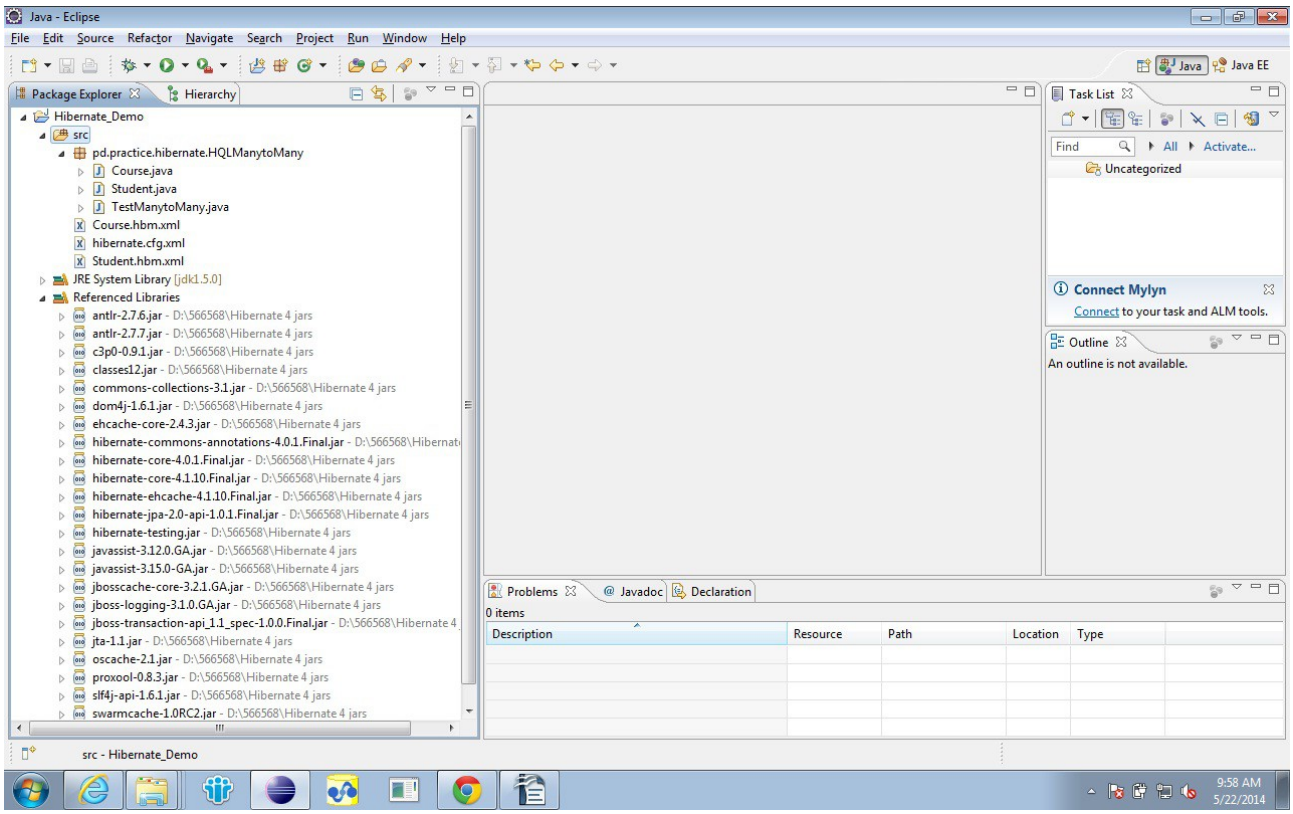


d) Browse through list of jar files in local directory

e) Add above mentioned list of jar files in the build path



f) Create sub pakages in src folder /Add hibernate.cfg.xml file in the class path of Hibernate_Demo Project(Just copy paste hibernate.cfg.xml file in src folder)



4 Important XML Tags of hibernate.cfg.xml

- ⇧ `<property name="hibernate.connection.driver_class">oracle.jdbc.driver.OracleDriver</property>`

The above tag points to the Oracle Driver class is required to execute the said Hibernate Project which exist in ***classes12.jar*** file

- ⇧ `<property name="hibernate.connection.url">jdbc:oracle:thin:@172.26.132.40:1521:ORCLILP</property>`

The above tag holds the connecting URL along with port no and schema name to which the Hibernate Project will connect.

- ⇧ `<property name="hibernate.connection.username">a109core</property>`
- ⇧ `<property name="hibernate.connection.password">a109core</property>`

The above two tag hold the database credentials required to connect to the underlying database.

- ⇧ `<property name="hibernate.dialect">org.hibernate.dialect.Oracle10gDialect</property>`

The above tag explains the kind of dialect that the project is going to use to execute this project. Here it indicates the dialect will be of Oracle10g specific.

- ```
  <property name="show_sql">true</property>
  ```

The above tag indicates to the Hibernate application to log/show the sql queries generated by the Hibernate framework in the Eclipse's console logs. This is very helpful to the developer if he/she wants to view the queries.

- ```
  <property name="hbm2ddl.auto">create</property>
  ```

This property when set to "create" will drop all the database tables and recreate them everytime the application is executed from the start. The value of this property generally depends on the enivironment.

While testing the hibernate application, this value needs to be changed to update as given below

- ```
  <property name="hbm2ddl.auto">update</property>
  ```

5 Caching Mechanism

- ```
  <property
  name="hibernate.cache.provider_class">org.hibernate.cache.EHCacheProvider<
  /property>
  ```
- ```
  <property
  name="hibernate.cache.region.factory_class">org.hibernate.cache.ehcache.Eh
  CacheRegionFactory</property>
  ```

The above tags is used to implement Caching Mechanism within Hibernate. When configured hibernate will implement a second-level cache. Also Hibernate will force to choose any one of several cache providers for the entire application. Second-level cache will cache data accross sessions. Session level cache provides the first-level cache.

- ```
  <mapping resource="Course.hbm.xml" />
  <mapping class="pd.practice.hibernate.HQLManytoMany.Course" />
  <mapping resource="Student.hbm.xml" />
  <mapping class="pd.practice.hibernate.HQLManytoMany.Student" />
  ```

The above tags are used to indicate the POJOs mapping with Database Entity which is further explained in ORM Mapping Topic. The ORM mappings contain the Java class(POJO) to database table mapping details.

6 Sample hibernate.cfg.xml file

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE hibernate-configuration PUBLIC
"-//Hibernate/Hibernate Configuration DTD 3.0//EN"
"http://www.hibernate.org/dtd/hibernate-configuration-3.0.dtd">
```

```
<hibernate-configuration>
	<session-factory>

		<property
name="hibernate.connection.driver_class">oracle.jdbc.driver.OracleDriver</property>
		<property
name="hibernate.connection.url">jdbc:oracle:thin:@172.26.132.40:1521:ORCLILP</property>
		<property name="hibernate.connection.username">a109core</property>
		<property name="hibernate.connection.password">a109core</property>
		<property
name="hibernate.dialect">org.hibernate.dialect.Oracle10gDialect</property>
		<property name="show_sql">true</property>
		<property name="hibernate.temp.use_jdbc_metadata_defaults">false</property>

		<property name="hbm2ddl.auto">create</property>
		<property name="format_sql">true</property>
		<property name="use_sql_comments">true</property>


		<property
name="hibernate.cache.provider_class">org.hibernate.cache.EHCacheProvider</property>

		<property
name="hibernate.cache.region.factory_class">org.hibernate.cache.ehcache.EhCacheRegionFactory
</property>

		<mapping resource="Course.hbm.xml" />
		<mapping class="pd.practice.hibernate.HQLManytoMany.Course" />
		<mapping resource="Student.hbm.xml" />
		<mapping class="pd.practice.hibernate.HQLManytoMany.Student" />

</session-factory>
</hibernate-configuration>
```